

Lab 3 Final Report

Unit 5 - Integrated E2E, visual evidence, and release integration

Local Evidence Draft

Status: local integrated evidence is verified. PR 5 review, merge, Issue #40 closure, and the separate release PR remain pending real GitHub events.

Item	Value
Branch	feature/lab3-integration-evidence
Implementation commit	4d4f114ab4224844584f59296e368de58d073fb9
Browser verification	9 Playwright tests passed; 12 screenshots inspected
Database verification	Migration, status, seed, and read-only count checks passed
Lab 2	Separate completed baseline unchanged

This PDF records local evidence only. It does not invent a PR number, review, approval, merge, or release event.

Answer Part 1 - Scope and objectives

Lab 3 integrates the completed authentication, Requester, IT Staff, and Administrator workflows; verifies them with API, UI, and browser checks; records responsive visual evidence; and prepares a reviewed release from lab3-staging to main.

The Unit 5 implementation is on feature/lab3-integration-evidence at commit `4d4f114ab4224844584f59296e368de58d073fb9`. Lab 2 remains a separate completed baseline and was not modified in this worktree.

The approved GitHub sequence remains exactly one Issue and one feature PR at a time for Unit 5: link Issue #40 through Development, request the friend review, reply to every comment, wait for approval, and have the reviewer merge PR 5 into lab3-staging before the separate release PR to main.

Answer Part 2 - Architecture and data model

The server remains the TypeScript/Express/Prisma service backed by PostgreSQL. The client remains the React/Vite application. Lab 3 adds the root Playwright entry point while keeping server and client unit tests in their existing package boundaries.

Docker PostgreSQL container toktickit-postgres was available for final verification. Migration 20260919000000_lab3_auth_requester was deployed successfully, prisma migrate status reported an up-to-date schema, and the seed was run twice successfully with stable 10-user/3-ticket seed results.

After the final browser run, a read-only count reported 11 users, 16 tickets, 4 public comments, and 3 internal notes. The additional records are from the integrated E2E flow.

Answer Part 3 - API, ownership, and validation

The server Lab 3 suite passed 22 tests across seven files, covering CSRF-safe authentication, password-change gating, authorization, migration regression, staff queue/detail operations, comments and notes, and Administrator user management.

The client suite passed 11 tests across five files. The browser suite passed the integrated Requester, IT Staff, and Administrator flows, including the Requester create/comment/resolution path and the staff/admin role boundaries.

The tested safety rules include separate public/internal content, role restrictions, and protection against self-deactivation or removal of the last active Administrator.

Answer Part 4 - UI, responsive behavior, and accessibility

The Unit 5 E2E definitions cover the Requester shell and regression flow, IT Staff Queue and Ticket Detail, and Administrator User Management. The shared fixture captures each scenario at 1280 x 900, 900 x 900, and 390 x 844.

All 12 PNGs were generated by the passing browser run and visually inspected. They are readable at all three sizes, with no visible clipped controls. The staff detail captures visibly keep Public Comments and Internal Notes separate.

The visual record does not claim a full keyboard or automated overflow audit; those items remain partially evidenced.

Check	Status
Desktop, tablet, and mobile layout	Passed - all 12 captures inspected
No apparent clipping	Partially evidenced - visual inspection passed; no automated overflow assertion
Keyboard/focus and role navigation	Partially evidenced - functional coverage passed; no full accessibility audit
Screenshot readability/caption accuracy	Passed

Answer Part 5 - Acceptance-criterion traceability

Criteria	Coverage	Status
AC-01 to AC-05	Auth API/UI and authentication.spec.ts	Passed
AC-06 to AC-10	Requester API/UI and authentication.spec.ts	Passed
AC-11 to AC-17	Staff API/UI and staff-ticket-flow.spec.ts	Passed
AC-18 to AC-25	Admin API/UI and user-administration.spec.ts	Passed
AC-26	Migration regression, Prisma, deploy/status, seed	Passed
AC-27	Responsive captures, visual/style/accessibility evidence	Partially evidenced - no dedicated keyboard/overflow audit

The detailed test matrix is in [docs/lab-03/tests.md](#). The figure links and captions are in [docs/lab-03/screenshot-index.md](#).

Answer Part 6 - Executed tests, builds, and database checks

Command/check	Result
server: npm test -- --run tests/lab-03	Passed - 7 files, 22 tests
server: npm run build	Passed
Prisma validate	Passed
Prisma migrate deploy	Passed - Lab 3 migration applied
Prisma migrate status	Passed - schema up to date
Prisma seed twice	Passed - 10 users and 3 tickets on both runs
client: npm test -- --run tests/lab-03	Passed - 5 files, 11 tests
client: npm run build	Passed - Vite transformed 31 modules
npx playwright test e2e/lab-03	Passed - 9 tests, 1 worker, 15.0s
Read-only final DB count	Passed - 11 users, 16 tickets, 4 comments, 3 notes
git diff --check	Run again after final evidence files are staged

The supporting command output and evidence labels are retained in docs/lab-03/release-evidence.md.

Answer Part 7 - Screenshot index and visual audit

Figures 1 through 12 are real PNG files generated by the passing Playwright run and were inspected directly. The same figure numbers and filenames are recorded in docs/lab-03/screenshot-index.md.

Figure	File	Viewport	Caption
1	artifacts/lab-03/screenshots/authentication/requester-regression/desktop-1280x900-my-tickets.png	1280 x 900	Requester My Tickets
2	artifacts/lab-03/screenshots/authentication/requester-regression/tablet-900x900-my-tickets.png	900 x 900	Requester My Tickets
3	artifacts/lab-03/screenshots/authentication/requester-regression/mobile-390x844-my-tickets.png	390 x 844	Requester My Tickets
4	artifacts/lab-03/screenshots/staff-queue/desktop-1280x900-queue.png	1280 x 900	IT Staff Queue
5	artifacts/lab-03/screenshots/staff-queue/tablet-900x900-queue.png	900 x 900	IT Staff Queue
6	artifacts/lab-03/screenshots/staff-queue/mobile-390x844-queue.png	390 x 844	IT Staff Queue
7	artifacts/lab-03/screenshots/staff-ticket-detail/desktop-1280x900-detail.png	1280 x 900	IT Staff Ticket Detail
8	artifacts/lab-03/screenshots/staff-ticket-detail/tablet-900x900-detail.png	900 x 900	IT Staff Ticket Detail
9	artifacts/lab-03/screenshots/staff-ticket-detail/mobile-390x844-detail.png	390 x 844	IT Staff Ticket Detail
10	artifacts/lab-03/screenshots/user-management/desktop-1280x900-list.png	1280 x 900	Administrator User Management
11	artifacts/lab-03/screenshots/user-management/tablet-900x900-list.png	900 x 900	Administrator User Management
12	artifacts/lab-03/screenshots/user-management/mobile-390x844-list.png	390 x 844	Administrator User Management

The updated PDF was rendered page by page; figure numbering, screenshot filenames, and the real GitHub links were checked.

Answer Part 8 - Peer review, merge, and issue tracking

The completed feature units are recorded by the following real GitHub links:

- [Lab 3 Issue 1 \(#36\)](#) and [PR 1 \(#41\)](#)
- [Lab 3 Issue 2 \(#37\)](#) and [PR 2 \(#42\)](#)
- [Lab 3 Issue 3 \(#38\)](#) and [PR 3 \(#43\)](#)
- [Lab 3 Issue 4 \(#39\)](#) and [PR 4 \(#44\)](#)
- [Lab 3 Issue 5 \(#40\)](#) is the approved Unit 5 issue.

PR 1 through PR 4 were reviewed and merged into lab3-staging. PR 5 has not been opened, so there is no PR 5 number or link to invent. Its Development-panel link, reviewer request, author reply, reviewer approval, merge, Issue #40 closure, and the separate reviewed lab3-staging to main release PR remain pending real GitHub events.

Answer Part 9 - Limitations and next actions

The remaining work is the controlled GitHub workflow, not unverified local implementation:

- Review the local evidence and keep the approved PR 5 body aligned with the actual files.
- Push feature/lab3-integration-evidence only when authorized.
- Open exactly PR 5 to lab3-staging, link Issue #40 through Development, request the designated review, and reply to every review comment.
- Wait for the friend approval; the reviewer, not the author, merges PR 5.
- Update the board and close Issue #40 only when the workflow permits it.
- Prepare and review the separate release PR from lab3-staging to main.

Evidence boundary

The local test, database, screenshot, and PDF results above are real and repeatable on this worktree. They do not assert any future GitHub action. The next external action requires the approved PR 5 sequence and the designated review.